

Constraint Satisfaction Problems

Lab 6

Exercise

Use the following pseudocode to complete the program [constraints_template.py](#).

Implement the recursive_backtrack function

Remember to **incorporate the other methods** already present in the file.

```
function RECURSIVE-BACKTRACKING(assignment, csp)  
  if assignment is complete then return assignment  
  var ← SELECT-UNASSIGNED-VARIABLE(VARIABLES[csp], assignment, csp)  
  for each value in ORDER-DOMAIN-VALUES(var, assignment, csp)  
    if value is consistent with assignment given CONSTRAINTS[csp]  
      add {var = value} to assignment  
      result ← RECURSIVE-BACKTRACKING(assignment, csp)  
      if result ≠ failure then return result  
      remove {var = value} from assignment  
  return failure
```

Exercise 2

Copy and modify the program from the exercise to use:

→ The map of South America (on the next slide)

→ 4 colors (red, green, blue and yellow)



Homework Challenge

Implement forward checking and arc consistency for the previous exercise.